

# IT23302-DATA STRUCTURES

NAME:- LALITH ANAND P.N

ROLL NO:-2025115014

1. **Write a C program to store n elements in an array using dynamic memory allocation and find the largest number.**

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, i, *a, max;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    a = (int *)malloc(n * sizeof(int));

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    max = a[0];
    for(i = 1; i < n; i++) {
        if(a[i] > max)
            max = a[i];
    }

    printf("Largest = %d", max);

    free(a);
    return 0;
}
```

OUTPUT:

## Output

```
Enter number of elements: 3
Enter elements:
12
34
18
Largest = 34
```

### 2. Print an array in reverse order using pointers.

```
#include <stdio.h>

int main() {
    int a[100], n, i;
    int *p;

    printf("Enter size: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    p = &a[n - 1];

    printf("Reverse array:\n");
    for(i = 0; i < n; i++) {
        printf("%d ", *p);
        p--;
    }

    return 0;
}
```

OUTPUT:

```
Enter size: 5
Enter elements:
18
7
06
08
27
Reverse array:
27 8 6 7 18

=== Code Execution Successful ===
```

**3. Write a C program to store information of students using structure and sort them by name.**

```
#include <stdio.h>
#include <string.h>

struct Student {
    int regno;
    char name[30];
    float mark;
};

int main() {
    struct Student s[50], temp;
    int n, i, j;

    printf("Enter number of students: ");
    scanf("%d", &n);

    // Input
    for(i = 0; i < n; i++) {
        printf("\nStudent %d\n", i + 1);

        printf("Reg No: ");
        scanf("%d", &s[i].regno);

        printf("Name: ");
        scanf("%s", s[i].name);
```

```

        printf("Mark: ");
        scanf("%f", &s[i].mark);
    }

    // Sort by name (Alphabetical)
    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(strcmp(s[i].name, s[j].name) > 0) {
                temp = s[i];
                s[i] = s[j];
                s[j] = temp;
            }
        }
    }

    // Display
    printf("\nStudents Sorted by Name\n");
    printf("RegNo\tName\tMark\n");

    for(i = 0; i < n; i++) {
        printf("%d\t%s\t%.2f\n",
            s[i].regno,
            s[i].name,
            s[i].mark);
    }

    return 0;
}

```

OUTPUT:

```

Enter number of students: 3

Student 1
Reg No: 44
Name: Dharshan
Mark: 100

Student 2
Reg No: 18
Name: Virat
Mark: 82

Student 3
Reg No: 07
Name: ERen
Mark: 46

Students Sorted by Name
RegNo  Name  Mark
44  Dharshan  100.00
7   ERen     46.00
18  Virat     82.00

=== Code Execution Successful ===

```

**4. Write a C program to add n distances given in inch–feet system using structures.**

```

#include <stdio.h>

struct Distance {
    int feet;
    int inch;
};

int main() {
    struct Distance d1, d2, sum;

    printf("Enter first distance (feet inch): ");
    scanf("%d%d", &d1.feet, &d1.inch);

    printf("Enter second distance (feet inch): ");
    scanf("%d%d", &d2.feet, &d2.inch);

```

```

sum.feet = d1.feet + d2.feet;
sum.inch = d1.inch + d2.inch;

if(sum.inch >= 12) {
    sum.feet++;
    sum.inch -= 12;
}

printf("Total Distance = %d feet %d inch", sum.feet, sum.inch);

return 0;
}

```

OUTPUT:

```

Enter first distance (feet inch): 4
5
Enter second distance (feet inch): 5
8
Total Distance = 10 feet 1 inch

=== Code Execution Successful ===

```

**5. Write a C program to add two complex numbers by passing structure to a function.**

```

#include <stdio.h>

struct Complex {
    int real;
    int imag;
};

struct Complex add(struct Complex a, struct Complex b) {
    struct Complex c;
    c.real = a.real + b.real;
    c.imag = a.imag + b.imag;
    return c;
}

int main() {
    struct Complex c1, c2, result;

```

```

printf("Enter first complex number (real imag): ");
scanf("%d%d", &c1.real, &c1.imag);

printf("Enter second complex number (real imag): ");
scanf("%d%d", &c2.real, &c2.imag);

result = add(c1, c2);

printf("Sum = %d + %di", result.real, result.imag);

return 0;
}

```

OUTPUT:

```

Enter first complex number (real imag): 5
8
Enter second complex number (real imag): 7
5
Sum = 12 + 13i

=== Code Execution Successful ===

```

## 6. Write a C program to maintain an Inventory Management System using Structure.

```

#include <stdio.h>

struct Item {
    int id;
    char name[30];
    int quantity;
    float price;
};

int main() {
    struct Item item[20];
    int n, i;
    float total;

    printf("Enter number of items: ");
    scanf("%d", &n);

    // Input item details
    for(i = 0; i < n; i++) {
        printf("\nItem %d\n", i + 1);
    }
}

```

```

    printf("Item ID: ");
    scanf("%d", &item[i].id);

    printf("Item Name: ");
    scanf("%s", item[i].name);

    printf("Quantity: ");
    scanf("%d", &item[i].quantity);

    printf("Price: ");
    scanf("%f", &item[i].price);
}

// Display inventory
printf("\n----- INVENTORY DETAILS ----- \n");
printf("ID\tName\tQty\tPrice\tTotal\n");

for(i = 0; i < n; i++) {
    total = item[i].quantity * item[i].price;

    printf("%d\t%s\t%d\t%.2f\t%.2f\n",
        item[i].id,
        item[i].name,
        item[i].quantity,
        item[i].price,
        total);
}

return 0;
}

```



OUTPUT:

```
Enter number of items: 3

Item 1
Item ID: 234
Item Name: RICE
Quantity: 3
Price: 60

Item 2
Item ID: 678
Item Name: WHEAT
Quantity: 2
Price: 40

Item 3
Item ID: 911
Item Name: MAGGIE
Quantity: 5
Price: 15

----- INVENTORY DETAILS -----
ID   Name      Qty  Price   Total
234  RICE       3    60.00   180.00
678  WHEAT      2    40.00   80.00
911  MAGGIE     5    15.00   75.00

=== Code Execution Successful ===
```

**7. Write a C program for decimal to binary, octal and hexadecimal conversion.**

```
#include <stdio.h>
```

```
int main() {
    int n;

    printf("Enter decimal number: ");
    scanf("%d", &n);

    printf("Binary: ");

    int temp = n, bin[32], i = 0;

    while(temp > 0) {
        bin[i++] = temp % 2;
        temp /= 2;
    }
}
```

```

while(i-->0)
    printf("%d", bin[i]);

printf("\nOctal: %o", n);
printf("\nHexadecimal: %X", n);

return 0;
}

```

OUTPUT:

```

Enter decimal number: 27.50
Binary: 11011
Octal: 33
Hexadecimal: 1B

=== Code Execution Successful ===

```

**8. Write a C program to calculate the difference between two time periods.**

```
#include <stdio.h>
```

```

struct Time {
    int hr;
    int min;
    int sec;
};

```

```

int main() {
    struct Time t1, t2, diff;

    printf("Enter first time (h m s): ");
    scanf("%d%d%d", &t1.hr, &t1.min, &t1.sec);

    printf("Enter second time (h m s): ");
    scanf("%d%d%d", &t2.hr, &t2.min, &t2.sec);

    if(t2.sec > t1.sec) {
        t1.sec += 60;
        t1.min--;
    }

    diff.sec = t1.sec - t2.sec;

    if(t2.min > t1.min) {
        t1.min += 60;
        t1.hr--;
    }
}

```

```

    }

    diff.min = t1.min - t2.min;
    diff.hr = t1.hr - t2.hr;

    printf("Difference = %d:%d:%d",
           diff.hr, diff.min, diff.sec);

    return 0;
}

```

OUTPUT:

```

Enter first time (h m s): 2
54
7
Enter second time (h m s): 4
45
18
Difference = -2:8:49

=== Code Execution Successful ===

```

**9. Write a C program to create an Employee Management System using structures.**

```

#include <stdio.h>

struct Employee {
    int id;
    char name[30];
    float salary;
};

int main() {
    struct Employee e[20];
    int n, i;

    printf("Enter number of employees: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("\nEmployee %d\n", i + 1);
        printf("ID: ");
        scanf("%d", &e[i].id);
        printf("Name: ");
    }
}

```

```

        scanf("%s", e[i].name);
        printf("Salary: ");
        scanf("%f", &e[i].salary);
    }

    printf("\nEmployee Details\n");
    for(i = 0; i < n; i++) {
        printf("%d %s %.2f\n",
            e[i].id, e[i].name, e[i].salary);
    }

    return 0;
}

```

OUTPUT:

```

Enter number of employees: 2

Employee 1
ID: 911
Name: DHARSHAN
Salary: 600000

Employee 2
ID: 108
Name: PRABHU
Salary: 420000

Employee Details
911 DHARSHAN 600000.00
108 PRABHU 420000.00

=== Code Execution Successful ===

```

**10. Write a C program to calculate area and perimeter of a rectangle and area of a triangle using function pointers.**

```
#include <stdio.h>
```

```

float rectArea(float l, float b) {
    return l * b;
}

```

```

float rectPerimeter(float l, float b) {
    return 2 * (l + b);
}

```

```

}

float triArea(float b, float h) {
    return 0.5 * b * h;
}

int main() {
    float l, b, h;
    float (*fp)(float, float);

    printf("Enter length and breadth: ");
    scanf("%f%f", &l, &b);

    fp = rectArea;
    printf("Rectangle Area = %.2f\n", fp(l, b));

    fp = rectPerimeter;
    printf("Rectangle Perimeter = %.2f\n", fp(l, b));

    printf("Enter triangle height: ");
    scanf("%f", &h);

    fp = triArea;
    printf("Triangle Area = %.2f", fp(b, h));

    return 0;
}

```

OUTPUT:

```

Enter length and breadth: 23
45
Rectangle Area = 1035.00
Rectangle Perimeter = 136.00
Enter triangle height: 34
Triangle Area = 765.00

=== Code Execution Successful ===

```

